

JEE MAIN REVISION MASTER: CHEMICAL EQUILIBRIUM

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Equilibrium Constants (K_c , K_p) & Le Chatelier's Principle	11 PYQs	Chemical Equilibrium

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Equilibrium Constants (K_c , K_p) & Le Chatelier's Principle	$K_p = K_c (RT)^{\Delta n_g}$, $\quad \text{Reversed Reaction: } K' = \frac{1}{K}$, $\quad n \times \text{Reaction: } K' = K^n$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Equilibrium Constants (K_c , K_p) & Le Chatelier's Principle (High Yield — 11 PYQs)

Core Principle: Equilibrium constant expression $K_p = K_c (RT)^{\Delta n_g}$. If reaction is multiplied by n , new constant is $K' = K^n$; if reversed, $K' = 1/K$. Le Chatelier's principle: Increasing pressure shifts equilibrium toward fewer gas moles; increasing temperature favours endothermic direction. Addition of inert gas at constant volume has NO effect on equilibrium.

■ **Caution / JEE Pitfall:** A catalyst increases the speed of both forward and reverse reactions equally; it does NOT alter the equilibrium constant or the yield of products!

■ **Speed Trick:** Addition of an inert gas at constant PRESSURE shifts equilibrium to the side with MORE moles of gas ($\Delta n_g > 0$ shifts forward).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "For the following gaseous reversible reaction: $3\text{A}_{(g)} + \text{B}_{(g)} \rightleftharpoons \text{A}_{(3)}\text{B}_{(g)}$ $\Delta_r H = -q$..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "For the reversible reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ When the partial pressure is measured i..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "The equilibrium constant at 298 K for the reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ is 100. If the initial concentrations of all the four speci..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

For the following gaseous reversible reaction: $3A_{(g)} + B_{(g)} \rightleftharpoons A_3B_{(g)}$ ($\Delta_r H = -q \text{ kJ}$). The amount of product $A_3B_{(g)}$ is affected by ____

- (A) Temperature alone
- (B) Pressure alone
- (C) Both temperature and pressure
- (D) Temperature, pressure and catalyst

Correct Answer: None

Solution Analysis: For the reaction $3A_{(g)} + B_{(g)} \rightleftharpoons A_3B_{(g)}$ the forward reaction is exothermic because $\Delta_r H = -q \text{ kJ}$. So, heat is released in the forward direction. If temperature is increased, the equilibrium shifts in the backward direction to absorb heat. So the amount of A_3B decreases. If temperature is decreased, the equilibrium shifts in the forward direction. So the amount of A_3B increases. Now look at pressure. On the reactant side, the number of gaseous moles is $3+1=4$. On the product side, the number